

Gelucire® 48/16: a safe wetting agent for tablet. A case study with fenofibrate

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INTRODUCTION

Gelucire® 48/16, a water-soluble surfactant, is known for its wetting and bioavailability enhancing capacity in the formulation of fenofibrate¹. With its high HLB (16) and its physical properties, Gelucire® 48/16 is a safe wetting agent that could be formulated into tablet.

This study aims to propose an alternative to sodium lauryl sulfate (SLS) used in the marketed reference, Lipanthyl.

MATERIALS AND METHODS

Gelucire® 48/16 (polyoxyl-32 stearate (type I) EP/NF), micronized fenofibrate, lactose monohydrate, microcrystalline cellulose, povidone K30, sodium carboxymethylcellulose, cross-linked PVP, silicon, colloidal silicon dioxide and magnesium stearate were used for the tablet formulation. The marketed product Lipanthyl was used as a reference. Gelucire® 48/16 pellets were ground (Quadro L1A) to obtain a powder form. The drug, surfactant, fillers and binder were poured in a mixer-granulator (Diosna P1-6). The granules were dried at 35°C for 28 min in an air bed fluidizer (Glatt, GPCG1.1) then sieved.

All other excipients were added successively and blended (15 min at 62 rpm) in a Turbula mixer (Bachofen AG, T2C) and lubricated 1.5 min at 62 rpm. Compression was carried out using a single station press (Korsch EK0) and oblong tablet punch (16*9*9 mm). Fenofibrate release from tablets was measured in 1000 mL buffer medium containing 0.72% of SLS, using a USP II paddle apparatus (Hanson) at 37 °C and 50 rpm. Samples were withdrawn up to 2 hours and analyzed using a UV-spectrophotometer at 285 nm (Agilent 8454).

RESULTS

A solubility screening showed that 10% w/w Gelucire® 48/16 in the tablet was required to solubilize the 160 mg therapeutic dose.

Since the drug is micronized, the main challenge was to optimize the flowability of the powder to ensure a homogeneous filling into the compression press. Diluent was hence carefully screened and selected. Sticking problem occurred during tableting. Hence, lubricant concentration and premixing process were optimized. As disintegration time was too long, different disintegrants at different ratios were evaluated: in this case, internal and external disintegrants were mandatory to conform to pharmacopeia specifications. The composition of the optimized formulation is given in Table 1.

Table 1. Fenofibrate tablet optimized composition.

Excipient	Chemical name	Function	%	
Pharmatose 200M	Lactose monohydrate	Diluent	23.1	Intra-granular 71.5%
Fenofibrate micronized	Fenofibrate micronized	Drug	23.5	
Avicel PH 102	Microcrystalline cellulose	Diluent	10.7	
Kollidon 30	Povidone K30	Binder	2.1	
Ac-Di-Sol SD-711	Croscarmellose sodium	Disintegrant	2.1	
Ground Gelucire® 48/16	PEG-32 Palmitostearate	Wetting agent	10.0	Extra-granular 28.5%
Prosolv SMCC HD90	Silicified Microcrystalline Cellulose	Diluent	18.5	
Kollidon CL	Cross-linked PVP Silicon	Disintegrant	5.0	
Aerosil 200 PHARMA	Silicon dioxide	Glidant	1.0	
Talc	Talcum	Glidant	3.0	
Magnesium Stearate	Magnesium stearate	Lubricant	1.0	

The obtained tablets had constant weight (691 mg ± 2), friability below 0.14%, hardness of 111.0 ± 3.9 N and a maximum disintegration time of 6 min 45 s.

The release of fenofibrate was faster within the first twenty minutes for Gelucire® 48/16-based tablets. The robustness was improved with a reduced standard deviation. Tablets were stable for 12 months at 25 °C/60% HR and 6 months at 40 °C/75% HR (Figure 1), with similarity F2 factor of 75 and 90 respectively.

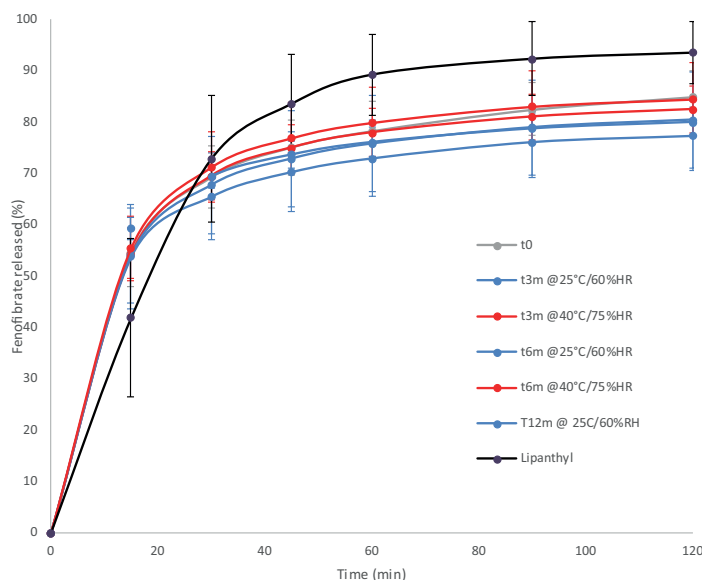


Figure 1. Stable drug release of Gelucire® 48/16 based tablet of fenofibrate.

CONCLUSION

The micronized fenofibrate tablets were successfully formulated with Gelucire® 48/16 as wetting agent, and proven robust and stable over time. This study highlights the benefits and possibility of using Gelucire® 48/16 to enhance the solubility and wettability of fenofibrate in oral solid dosage forms.

REFERENCES

¹ Gattefosse, available from <https://www.gattefosse.com/fr/pharmaceuticals-products/gelucire-4816> (last consult: 2022/02/15)